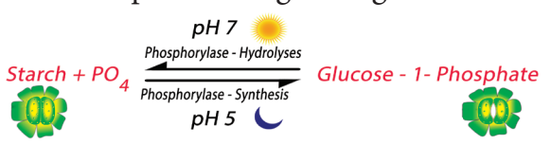
2. The guard cells already possess much amount of stored sugars.

2. Starch – Sugar Interconversion theory

i. According to **Lloyd** (1908), turgidity of guard cell depends on interconversion, of starch and sugar. It was supported by **Loftfield** (1921) as he found guard cells containing sugar during the daytime when they are open and starch during the night when they are closed.

ii. **Sayre** (1920) observed that the opening and closing of stomata depends upon change in pH of guard cells. According to him stomata open at high pH during day time and become closed at low pH at night. Utilization of CO_2 by photosynthesis during light period causes an increase in pH resulting in the conversion of starch to sugar. Sugar increase in cell favours endosmosis and increases the turgor pressure which leads to opening of stomata. Likewise, accumulation of CO_2 in cells during night decrease the pH level resulting in the conversion of sugar to starch. Starch decreases the turgor pressure of guard cell and stomata close.

iii. The discovery of enzyme **phosphorylase** in guard cells by **Hanes** (1940) greatly supports the starch-sugar interconversion theory. The enzyme *phosphorylase* hydrolyses starch into sugar and high pH followed by endosmosis and the opening of stomata during light. The *vice versa* takes place during the night.



iv. **Steward** (1964) proposed a slightly modified scheme of starch-sugar interconversion theory. According to him, Glucose-1-phosphate is osmotically inactive. Removal of phosphate from Glucose-1-phosphate converts to Glucose which is osmotically active and increases the concentration of guard cell leading to opening of stomata (Figure 11.15).

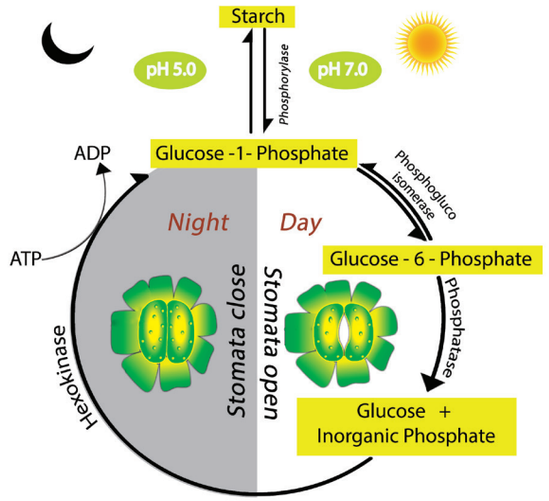


Figure 11.15: Steward Scheme

Objections to Starch-sugar interconversion theory

i. In monocots, guard cell does not have starch.

ii. There is no evidence to show the presence of sugar at a time when starch disappears and stomata open.

iii. It fails to explain the drastic change in pH from 5 to 7 by change of CO_2.

3. Theory of K^+ transport

This theory was proposed by **Levit** (1974) and elaborated by **Raschke** (1975). According to this theory, the following steps are involved in the stomatal opening:

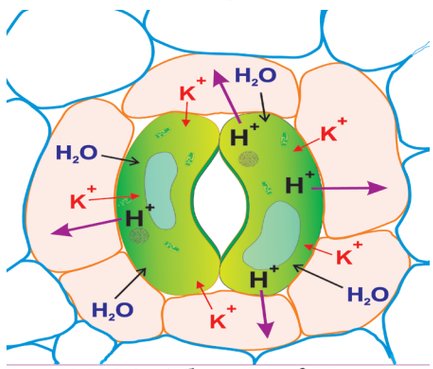


Figure 11.16: Theory of K^+ transport

Opening of stomata

In light

i. In guard cell, starch is converted into organic acid (malic acid).

ii. Malic acid in guard cell dissociates to malate anion and proton (H^+).

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